Supplementary file 1 – Search strings and references with reasons for rejection during the screening process.

Search strings

Source	**Search string**
PUBMED	((((ketamine[MeSH Terms]) AND midazolam[MeSH Terms]) AND preanesthetic medication[MeSH Terms])) Filters: Clinical Trial; Humans; Child: birth-18 years"
CENTRAL	"midazolam" AND "ketamine" in Record Title AND ("preanesthetic" OR"preanesthesia" OR "premedication") in Title Abstract Keyword AND "oral" NOT "intranasal" NOT "transmucosal" in Title Abstract Keyword AND ("children" OR "paediatric") in Title Abstract Keyword AND "randomised control trial" in Publication Type - (Word variations have been searched)
Embase	children:ab AND 'oral midazolam' AND 'ketamine'/exp AND 'midazolam'/mj AND 'sedation'/exp AND ('randomized clini' OR 'premedication'/exp)
Web of science	TS=(midazolam AND ketamine) AND TS = (preanesthesia medication OR preanesthetic medication) AND TS=child* *Indexes=SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, ESCI Timespan=All years*
Scholar Google	allintitle: midazolam ketamine children oral preanesthetic OR premedication OR oral -intranasal -transmucosal -intramuscular -rectal

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1. *Astuto M, Disma N, Crimi E.Two doses of oral ketamine, given with midazolam, for premedication in children.Minerva Anestesiol. 2002 Jul-Aug;68(7-8):593-8.*	*Included in meta-analyses*
2. *Darlong V, Shende D, Singh M, Garg R, Pandey R, Punj J.Low- versus high-dose combination of midazolam-ketamine for oral premedication in children for ophthalmologic surgeries.Singapore Med J. 2011 Jul;52(7):512-6. Erratum in: Singapore Med J. 2011 Sep;52(9):704.*	*Included in meta-analyses*
3. *Darlong V, Shende D, Subramanyam MS, Sunder R, Naik A.Oral ketamine or midazolam or low dose combination for premedication in children.Anaesth Intensive Care. 2004 Apr;32(2):246-9.*	*Included in meta-analyses*
4. *Foroutan A, Yazdanian F, Panahipour AA, Meraji SM, Givtaj N, Jafari AR. ORAL PREMEDICATION FOR PEDIATRIC CARDIAC SURGERY: A COMPARISON OF MIDAZOLAM, KETAMINE AND MIDAZOLAM PLUS KETAMINE. .IRANIAN HEART JOURNAL (IHJ). 2008;8:17-23.*	*Included in meta-analyses*
5. *Funk W., Jakob W., Riedl T., Taeger K.Oral preanaesthetic medication for children: Double-blind randomized study of a combination of midazolam and ketamine vs midazolam or ketamine alone.British Journal of Anaesthesia (2000) 84:3 (335-340). Date of Publication: March 2000*	*Included in meta-analyses*
6. *Ghai B, Grandhe RP, Kumar A, Chari P.Comparative evaluation of midazolam and ketamine with midazolam alone as oral premedication.Paediatr Anaesth. 2005 Jul;15(7):554-9.*	*Included in meta-analyses*
7. *Hasani, M; Comparison Of Oral Premedication With Combination Of Midazolam With Ketamine Vs Midazolam Ketamine Alone In Children Children Medical Center (year 2000).Tehran University Medical Journal TUMS Publications.2002;60:423-428*	*Included in meta-analyses*
8. *Jain K., Ghai B., Saxena A.K., Saini D., Khandelwal N.Efficacy of two oral premedicants: Midazolam or a low-dose combination of midazolam-ketamine for reducing stress during intravenous cannulation in children undergoing CT imaging.Paediatric Anaesthesia (2010) 20:4 (330-337). Date of Publication: April 2010*	*Included in meta-analyses*

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9. *Kulikov AS, Sorokin VS, Lubnin AIu.[Oral premedication with midasolam and ketamine in children with neurosurgical diseases].Anesteziol Reanimatol. 2010 Jan-Feb;(1):6-10. Russian.*	*Included in meta-analyses*
10. *Kumar A., Shah Z.A., Anuradha, Garg R., Nath M.P.Comparative evaluation of ketamine, midazolam and combination of both as oral premedicants in children.Journal of Anaesthesiology Clinical Pharmacology (2009) 25:4 (449-453). Date of Publication: October 2009*	*Included in meta-analyses*
11. *Lin YC, Moynihan RJ, Hackel A. A comparison of oral midazolam, oral ketamine and oral midazolam combined with ketamine as preanesthetic medication for pediatric outpatients. .Anesthesiology. 1993;70:A1177*	*Included in meta-analyses*
12. *Magar J, Kotwani MBK, Kotak S. A double blinded comparative study of oral premedication in children with midazolam alone or in combination with ketamine. .International Journal of Contemporary Pediatrics; Vol 3, No 3 (2016): July-September 2016. 2016.*	*Included in meta-analyses*
13. *Majidinejad S., Taherian K., Esmailian M., Khazaei M., Samaie V.Oral midazolam-ketamine versus midazolam alone for procedural sedation of children undergoing computed tomography; a randomized clinical trial.Emergency (2015) 3:2 (64-69). Date of Publication: 2015*	*Included in meta-analyses*
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16. *Ramakrishna R, Hemanth KJ2, Sunil BV3,*, Neeta S4, Sonal Bhat5 Oral premedication in children: A randomized study of a combination of oral midazolam, ketamine with atropine vs midazolam and atropine vs ketamine and atropine . Indian Journal of Clinical Anaesthesia, April-June, 2018;5(2):261-265*	*Included in meta-analyses*
17. *Sajedi P, Aghadavoudi O, Salimi-Jazi F. Oral Midazolam Alone or in Combination with Ketamine as Oral Premedication in Pediatric Ophthalmologic Surgeries. .J Isfahan Med Sch 2014; 31(262): 1901-9*	*Included in meta-analyses*
18. *Sathyan, N; Comparative Evaluation of Two Doses of Ketamine with Midazolam, Ketamine alone and Midazolam alone as Oral Premedication in Children: A Study of 100 cases..2006;:*	*Included in meta-analyses*
19. *Walia C, Pankaj; Manjunath Prabhu, Daniel Thomas; Oral premedication in children: Comparison of combination of midazolam-ketamine and oral midazolam-A Randomised trial.Indian Journal of Clinical Anaesthesia.2018;5:249-254*	*Included in meta-analyses*
20. *Warner DL, Cabaret J, Velling D. Ketamine plus midazolam, a most effective paediatric oral premedicant. .Paediatr Anaesth. 1995;5(5):293-295. doi:10.1111/j.1460-9592.1995.tb00307.x*	*Included in meta-analyses*

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22. *Shah S., Shah S., Apuya J., Gopalakrishnan S., Martin T.Combination of oral ketamine and midazolam as a premedication for a severely autistic and combative patient.Journal of Anesthesia (2009) 23:1 (126-128). Date of Publication: 2009*	*Rejected after reading the abstract: case report*
23. *Done V., Kotha R., Vasa A.A.K., Sahana S., Jadadoddi R.K., Bezawada S.A comparison of the effectiveness of oral midazolam – N(2)O versus oral ketamine – N(2)O in pediatric patients-an in–vivo study.Journal of Clinical and Diagnostic Research (2016) 10:4 (ZC45-ZC48) Article Number: 111. Date of Publication: 1 Apr 2016*	*Rejected after reading the abstract: concomitant N2O administration*
24. *Rahsan Karayazili, Isil Ozkocak, Hilal Ayoglu, Serhan Yurtlu, Dilek Okyay and Volkan HanciComparison of the effects of oral midazolam, ketamine and tramadol on postoperative agitation related to sevoflurane in children.The Medical Bulletin of Haseki.2010;:146-153*	*Rejected after reading the abstract: does no include outcomes of interest*
25. *Bhatnagar S., Mishra S., Gupta M., Srikanti M., Mondol A., Diwedi A.Efficacy and safety of a mixture of ketamine, midazolam and atropine for procedural sedation in paediatric oncology: A randomised study of oral versus intramuscular route.Journal of Paediatrics and Child Health (2008) 44:4 (201-204). Date of Publication: April 2008*	*Rejected after reading the abstract: I.M administration*

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34. *Alderson PJ, Lerman J.Oral premedication for paediatric ambulatory anaesthesia: a comparison of midazolam and ketamine.Can J Anaesth. 1994 Mar;41(3):221-6.*	*Rejected after reading the abstract: no MIKE group included*
35. *F Motamed, Y Aminpour, H Hashemian, AE Soltani, M Najafi, F FarahmandMidazolam-ketamine combination for moderate sedation in upper GI endoscopy.Journal of pediatric gastroenterology and nutrition, 2012, 54(3), 422-426 | added to CENTRAL: 28 February 2019 | 2019 Issue 2*	*Rejected after reading the abstract: no MIKE group included*
36. *Pattnaik, SK (Pattnaik, Shaswat Kumar); Samal, S (Samal, Soumya); Jena, SK (Jena, Saubhagya Kumar)COMPARATIVE STUDY OF EFFECTS OF MIDAZOLAM AND KETAMINE AS ORAL PREMEDICATION FOR PAEDIATRIC ANAESTHESIA.JOURNAL OF EVOLUTION OF MEDICAL AND DENTAL SCIENCES-JEMDS Volume: 5 Issue: 54 Pages: 3611-3615 DOI: 10.14260/jemds/2016/832 Published: JUL 7 2016*	*Rejected after reading the abstract: no MIKE group included*
37. *Rubinstein O., Barkan S., Breitbart R., Berkovitch S., Toledano M., Weiser G., Karadi N., Nassi A., Kozer E.Efficacy of oral ketamine compared to midazolam for sedation of children undergoing laceration repair.Medicine (United States) (2016) 95:26 Article Number: e3984. Date of Publication: 1 Jun 2016*	*Rejected after reading the abstract: no MIKE group included*

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39. *Valdovinos E.H., Soto P.C., Solache L.B.G.Level of sedation in using midazolam vs oral ketamine in pediatric preanesthetic medication.Anestesia en Mexico (2008) 20:3 (117-121). Date of Publication: September-December 2008*	*Rejected after reading the abstract: no MIKE group included*
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41. *Roelofse, JA (Roelofse, JA); Joubert, JJD (Joubert, JJD); Roelofse, PGR (Roelofse, PGR)A double-blind randomized comparison of midazolam alone and midazolam combined with ketamine for sedation of pediatric dental patients.JOURNAL OF ORAL AND MAXILLOFACIAL SURGERY Volume: 54 Issue: 7 Pages: 838-844 DOI: 10.1016/S0278-2391(96)90531-5 Published: JUL 1996*	*Rejected after reading the abstract: rectal administration*
42. *O'Hara D., Ganeshalingam K., Gerrish H., Richardson P.A 2 year experience of nurse led conscious sedation in paediatric burns.Burns (2014) 40:1 (48-53). Date of Publication: February 2014*	*Rejected after reading the abstract: retrospective case series*
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45. *Baygin, O (Baygin, Ozgul); Bodur, H (Bodur, Haluk); Isik, B (Isik, Berrin)Effectiveness of premedication agents administered prior to nitrous oxide/oxygen.EUROPEAN JOURNAL OF ANAESTHESIOLOGY Volume: 27 Issue: 4 Pages: 341-346 DOI: 10.1097/EJA.0b013e3283313cdd Published: APR 2010*	*Rejected after reading the full article: no outcomes of interest*
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48. *Khatavkar, Sonal S; Deshpande, Trupti P; Zayed, Sameer; Misal, Manoj Zingraji; Kate, Pranita; Comparison of oral Midazolam with Ketamine versus oral Midazolam as a premedication in children.International Journal of Pharmaceutical Chemistry and Analysis.2016;3:66-72*	*Rejected after reading the full article: outcomes reported as mean and standard deviation*

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50. *Senturk Z., Kuvaki B., Gokmen N., Arkan A., Usta A., Atay A.Comparison of oral diazepam, midazolam and midazolam-ketamin, premedications in pediatric patients.Turk Anesteziyoloji ve Reanimasyon (1997) 25:10 (438-446). Date of Publication: 1997*	*Rejected after reading the full article: outcomes reported as mean and standard deviation*
51. *Abdel-Ghaffar HS, Kamal SM, El Sherif FA, Mohamed SA.Comparison of nebulised dexmedetomidine, ketamine, or midazolam for premedication in preschool children undergoing bone marrow biopsy.Br J Anaesth. 2018 Aug;121(2):445-452. doi: 10.1016/j.bja.2018.03.039. Epub 2018 Jun 22.*	*rejected at title screening*
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57. *Arai, YCP (Arai, Young-Chang P.); Ueda, W (Ueda, Wasa); Ito, H (Ito, Hiroshi); Wakao, Y (Wakao, Yoshiko); Matsura, M (Matsura, Masayo); Komatsu, T (Komatsu, Toru)Maternal heart rate variability just before surgery significantly correlated with emergence behavior of children undergoing general anesthesia.PEDIATRIC ANESTHESIA Volume: 18 Issue: 2 Pages: 167-171 DOI: 10.1111/j.1460-9592.2007.02363.x Published: FEB 2008*	*rejected at title screening*
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60. *Bahetwar, SK (Bahetwar, S. K.); Pandey, RK (Pandey, R. K.); Saksena, AK (Saksena, A. K.); Chandra, G (Chandra, Girish)A Comparative Evaluation of Intranasal Midazolam, Ketamine and their Combination for Sedation of Young Uncooperative Pediatric Dental Patients: A Triple Blind Randomized Crossover Trial.JOURNAL OF CLINICAL PEDIATRIC DENTISTRY Volume: 35 Issue: 4 Pages: 415-420 DOI: 10.17796/jcpd.35.4.l43h3354705u2574 Published: SUM 2011*	*rejected at title screening*
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65. *Cingi E., Beebe D., Whitley C., Belani K.Preprocedural administration of midazolam-safety and efficacy in children with sanfilippo syndrome type a.Molecular Genetics and Metabolism (2014) 111:3 (292). Date of Publication: March 2014*	*rejected at title screening*
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72. *Golparvar M, Saghaei M, Sajedi P, Razavi SS.Paradoxical reaction following intravenous midazolam premedication in pediatric patients - a randomized placebo controlled trial of ketamine for rapid tranquilization.Paediatr Anaesth. 2004 Nov;14(11):924-30.*	*rejected at title screening*
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Summary table

Status	**n**
Retrieved references	**145**
After duplicates removal	**105**
Rejected at title screening	**52**
Rejected after reading the abstract: case report	**2**
Rejected after reading the abstract: concomitant N2O administration	**1**
Rejected after reading the abstract: does no include outcomes of interest	**1**
Rejected after reading the abstract: I.M administration	**3**
Rejected after reading the abstract: I.V. administration	**2**
Rejected after reading the abstract: ketamine as control	**1**
Rejected after reading the abstract: letter	**1**
Rejected after reading the abstract: no Midazolam control included	**2**
Rejected after reading the abstract: no MIKE group included	**7**
Rejected after reading the abstract: rectal administration	**1**
Rejected after reading the abstract: retrospective case series	**1**
Rejected after reading the full article: no MDZ control group	**1**
Rejected after reading the full article: no outcomes of interest	**4**
Rejected after reading the full article: outcomes reported as mean and standard deviation	**3**
Unavailable - authors and editors contacted without response	**3**
Included in meta-analyses	**20**